

Question number	Answer	Additional guidance	Mark
4(a)	An explanation that includes three of the following points: - dopamine cannot cross the blood brain barrier (1) - this drug ensures more L-dopa {is present in blood /can cross the brain barrier / reaches the brain} (1) - resulting in (a greater) increase in dopamine levels in the brain (1) - statement about why increased dopamine would help treat Parkinson's (1)	ACCEPT larger ignore large ACCEPT L-dopa (only) converted to dopamine in the brain e.g. transmission of electrical impulses to post synaptic neurones / increase in motor responses	(3)

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4(b)(i)	An explanation that includes the following points: • {control gene / genetic engineering process} has no effect (on L-dopa synthesis) (1) • MYB12 gene has {little / no} effect on L-dopa synthesis (1) • The CYP76 gene causes a {big/ 9mg} increase in concentration of L-dopa (because it codes for a protein involved in the production of L-dopa) (1) • combination of MYB12 and CYP76 produce largest concentration of L-dopa because {MYB12/ the transcription factor} increases production of CYP76 protein (1)	ACCEPT it increases it by 1.25 times/eq ACCEPT CYP76 gene gives {most / more / larger} L-dopa when used individually NOTE if they state CYP76>MYB12>control then award mp3	(4)

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4(b)(ii)	A description that includes the following points: • {the gene / CYP76} that converts tyrosine to L dopa is {extracted / isolated} (1) • use of same {endonuclease / restriction enzyme} to cut genetic material of plasmid (and isolated gene) (1) • method of insertion of vector into a {tomato/host} cell (1)	ACCEPT {isolate / extract / cut out / synthesise} the gene ACCEPT description of method of insertion e.g. use of {gene gun / heat shock/ liposomes / CRISPR} e.g. infect using {non-lytic virus / Agrobacterium tumefaciens}	(3)

